

BRAC University

CSE331 : Automata And Computability

SPRING 2025

Quiz - 5

SET - A

Marks :20

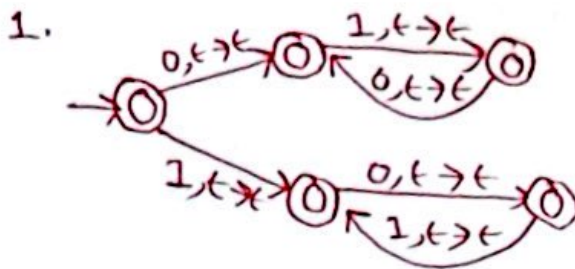
Duration: 20 mins

CO2	Name:	ID:	Sec:
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Draw pushdown automata for the following languages.

1. $L = \{w \in \{0, 1\}^* ; 0 \text{ and } 1 \text{ alternates in } w\}$ $010101 \dots, 101010 \dots$ [8]

2. $L = \{w \in \{0, 1, 2\}^* w = 0^i 1^j 2^k ; 2i+3j+3n = k, \text{ where } i, j, n \geq 0\}$ [12]

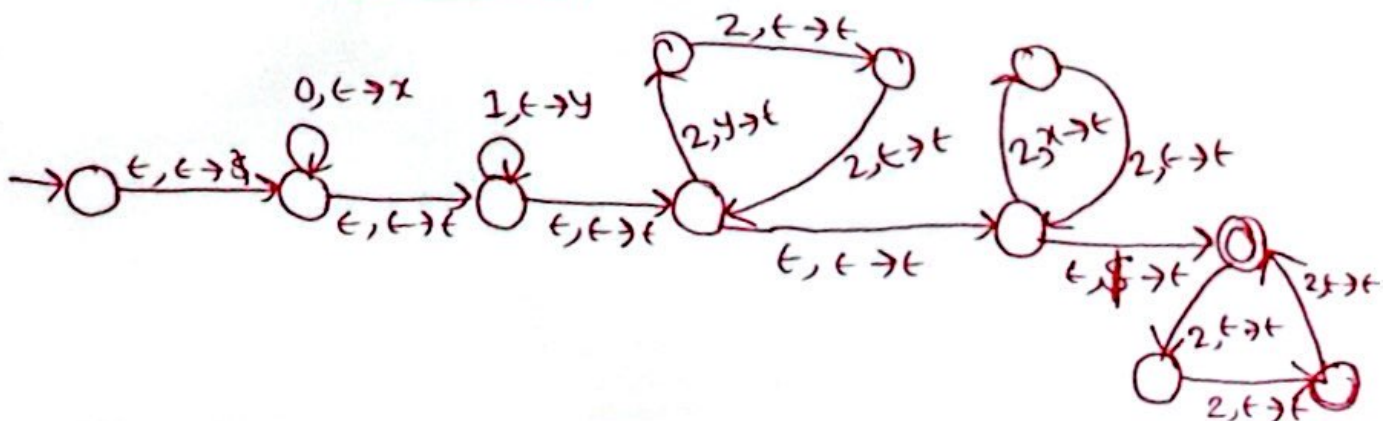


2.

$$0^i 1^j 2^k$$

$$= 0^i 1^j 2^{2i+3j+3n}$$

$$= 0^i \underbrace{1^j 2^{3j}}_{2^{3j}} 2^{2i} 2^{3n}$$



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SET - B

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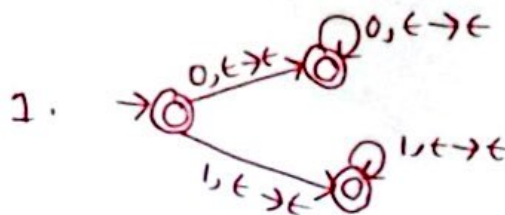
Duration: 20 mins

CO2	Name:	ID:	Sec:
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Draw pushdown automata for the following languages.

1. $L = \{ w \in \{0,1\}^* ; w \text{ neither have } 01 \text{ nor } 10 \text{ as substring} \}$, 000...11 [8]

2. $L = \{ w \in \{0, 1, 2\}^* w = 0^i 1^j 2^k ; 3i+2n+2k = j, \text{ where } i,k,n \geq 0 \}$ [12]



2.

$$0^i 1^j 2^k$$

$$= 0^i 1^{3i+2n+2k} 2^k$$

$$= \underbrace{0^i 1^{3i}}_{\text{part 1}} \underbrace{1^{2n}}_{\text{part 2}} \underbrace{1^{2k} 2^k}_{\text{part 3}}$$

